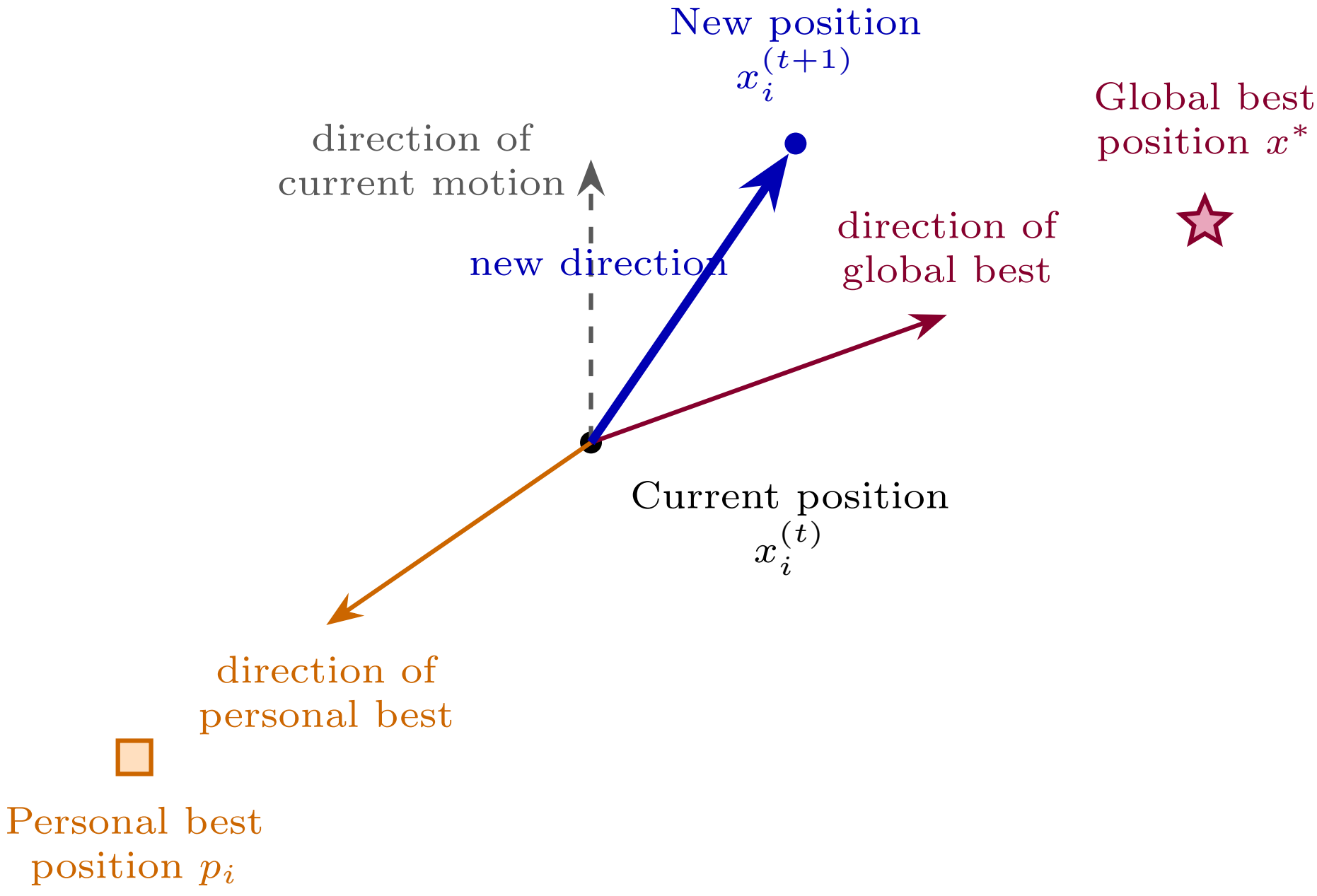
New position
$x_i^{(t+1)}$
Global best
position $x^*$
direction of
current motion
new direction
direction of
global best
Current position
$x_i^{(t)}$
direction of
personal best
Personal best
position $p_i$